

9 (long) Some decorative lights consist of banks of red (R) and yellow (Y) lights, with each bank of lights being turned on or off by a single output from a controller which is to be constructed from J-K bistables. When the lights are first switched on, they are to follow sequence 1 below:

$$\overline{R}\overline{Y}, R.Y, \overline{R}\overline{Y} \dots \quad (\text{sequence 1})$$

Pressing a button P (so $P = 1$) whilst in either state of sequence 1 will cause the lights to enter sequence 2 below, in the $R.\overline{Y}$ state:

$$R.\overline{Y}, \overline{R}.Y, R.\overline{Y} \dots \quad (\text{sequence 2})$$

Pressing button P again (so $P = 0$) whilst in either state of sequence 2 will cause the lights to enter sequence 1 again in the $\overline{R}\overline{Y}$ state.

- (a) Draw a state diagram for the controller. [8]
- (b) Determine the number of J-K bistables required to implement the system. [2]
- (c) Find the corresponding state transition table and hence determine the J-K inputs of the bistables in terms of the state of the system outputs and push button P . [10]
- (d) Draw a circuit diagram that implements the controller. It is essential that on applying power to the controller the lights start off in the $\overline{R}\overline{Y}$ state. [10]